Cells: Structure, Transport & Division

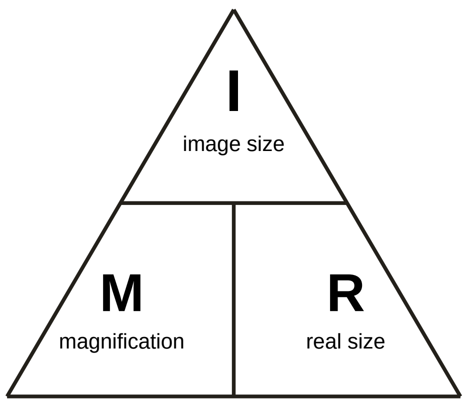
Quick reference sheet · Biology & Maths · keep handy for revision

1. Microscopy & magnification

An electron microscope has far higher resolving power than a light microscope (up to ×1,000,000 vs ×1,000), so it reveals much finer sub-cellular detail.

magnification = size of image ÷ size of real object or M = I / R

Image and real size must be in the **same units** before dividing. 1 mm = 1000 µm.



Cover **I** → M next to R → **I = M × R**. Cover **M** → I over R → **M = I / R**. Cover **R** → I over M → **R = I / M**.

2. Cell structures

Animal & plant cells are **eukaryotic** (genetic material in a nucleus). Bacterial cells are **prokaryotic** — smaller, with a single DNA loop and plasmids, not in a nucleus.

Structure	Animal	Plant	Bacterial	Function
Nucleus	✓	✓	—	Contains genetic material
Cytoplasm	✓	✓	✓	Most chemical reactions
Cell membrane	✓	✓	✓	Controls entry/exit
Mitochondria	✓	✓	—	Respiration (releases energy)
Ribosomes	✓	✓	✓	Protein synthesis
Cell wall	—	✓ (cellulose)	✓	Strengthens the cell
Chloroplasts	—	✓	—	Photosynthesis
Permanent vacuole	—	✓	—	Cell sap, keeps cell rigid
DNA loop & plasmids	—	—	✓	Genetic material (no nucleus)

3. Transport across membranes

All three processes move particles across the cell membrane — the difference is **what** moves, **which way**, and whether **energy** is needed. A "concentration gradient" just means one side has more particles of something than the other.

Process	What moves	Direction	Energy	Example
Diffusion	Particles	High → low conc.	No	O_2 alveoli → blood
Osmosis	Water only	High → low water conc., partially permeable membrane	No	Water into root hairs
Active transport	Dissolved substances	Low → high conc. (against gradient)	**Yes**	Mineral ions into root hairs

Diffusion — the **net** movement of particles from an area of **higher concentration** to an area of **lower concentration**, until the particles are evenly spread out. It happens because particles are always randomly moving, and needs **no energy** from the cell. Example: oxygen diffuses from the air in the alveoli (high O_2) into the blood (lower O_2); carbon dioxide diffuses the opposite way.

Osmosis — a **special case of diffusion for water only**: the net movement of water from a region of **higher water concentration** to a region of **lower water concentration**, through a **partially permeable membrane** (one that lets small water molecules through but not larger dissolved particles). No energy is needed. Example: water moves from the soil (dilute, lots of water) into root hair cells (more concentrated cell sap, less water).

Active transport — the movement of dissolved substances from a **lower** to a **higher** concentration — i.e. **against** the concentration gradient, the opposite way to diffusion. Because this doesn't happen naturally, the cell must supply **energy from respiration** to make it happen. Example: mineral ions (e.g. nitrate) are often more concentrated *inside* root hair cells than in the soil, so they can only get in via active transport — diffusion alone couldn't move them "uphill".

Exam tip: *for an "explain" question (like E3), always give the **direction** of movement first, say whether that is **with** or **against** the gradient, then link to **energy from respiration** if it's against the gradient. For a "define" question (E1/E2), state **what moves** and the **direction** — for osmosis, also mention "partially permeable membrane".*

4. Cell division: mitosis & meiosis

Human body cells have **46 chromosomes** (23 pairs) — thread-like structures made of DNA that carry genes. Before any division, the cell first copies (replicates) all its chromosomes and organelles, so it has everything needed to make new complete cells.

Mitosis produces **2 genetically identical** cells, each with the full 46 chromosomes — used for two reasons: **growth** (making an organism bigger by adding new cells) and **repair** (replacing damaged or worn-out cells, e.g. healing skin). This is the only type of division most body cells ever use.

Meiosis only happens in reproductive organs (ovaries/testes), to make **gametes** (sex cells: eggs and sperm). The cell divides **twice**, producing **4 cells**, each with **half** the chromosome number (23, one from each pair) — and each gamete is **genetically different** from the others because the chromosome pairs are shuffled before splitting. At **fertilisation**, an egg (23) and sperm (23) fuse to restore the full number (46) in the new cell, which then divides by mitosis to grow into an embryo.

	Mitosis	Meiosis
Purpose	Growth & repair	Making gametes
Divisions	One	Two

Cells produced	2	4
Chromosome no.	Same as parent (46)	Half the parent (23)
Genetically identical?	Yes	No — all different

Exam tip: *"state the number of chromosomes and explain why" (like F2) needs three things — the body-cell number (46), the gamete number (23),* ***and*** *the reason: meiosis halves the number so that fertilisation (joining two gametes) can restore 46 without doubling it every generation.*

5. Cell differentiation & stem cells

Early embryo cells are unspecialised **stem cells** — they can divide repeatedly and become almost any type of cell in the body. As the embryo develops, most cells **differentiate**: they change to become **specialised** for one particular job, developing a shape and sub-cellular structures suited to that function — e.g. a red blood cell loses its nucleus to carry more oxygen, a nerve cell grows a long fibre to carry electrical signals, a sperm cell grows a tail and packs in mitochondria for energy.

Differentiation is normally a **one-way process**: once a cell has specialised, it **cannot** change into a different cell type. This is why a nerve cell can never become a muscle cell, even though both came from the same original stem cell.

Most adult tissues keep a small number of **adult stem cells** (e.g. in bone marrow), which stay unspecialised and can divide to replace damaged or worn-out cells — but adult stem cells can usually only become a **limited range** of cell types (e.g. bone marrow stem cells make different blood cells). **Embryonic** stem cells are more versatile and can become **any** cell type, which is why scientists research them for treating diseases (e.g. repairing damaged nerves or growing new tissue) — though this raises ethical questions since it involves using embryos.